01. SELECT DISTINCT department
FROM students;
RESULTS ↓

Department
IT
HR
IT
Finance
HR

02. SELECT department_name,
AVG(age) AS average_age
FROM students
GROUP BY department name;
RESULTS ↓

Department	Avg Age
IT	20.5
HR	22
Finance	23

03. SELECT departments,
COUNT(students.name) AS count_students
FROM students-table
GROUP BY departments
HAVING COUNT(students) > 1;

RESULTS ↓

Department	Student Count
IT	2
HR	2

04. SELECT students_name, department
FROM students-table
WHERE age BETWEEN 21 and 23
RESULTS ↓

Student_id	Name	age	Department
2	Bob	22	HR
5	Eve	22	HR

05. SELECT students_name,
FROM student-table
WHERE department = 'IT' OR 'HR' AND
age > 21
RESULTS

Student_id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

06. SELECT department
SUM (credits) AS total_credits
FROM courses-table
GROUP BY departments
HAVING ~~COUNT~~ SUM (credits) > 5;

RESULTS ↓

department	total_Credits
IT	11

07. SELECT course_name, department
FROM courses-table
WHERE credits < 4
RESULTS ↓

course_id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08 SELECT course-name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
RESULTS ↓

course_id	course_name	credits
102	~~SQL Basics~~ PYTHON	4
103	Data Science	4
101	SQL Basics	3
105	Statistics	3
104	Excel	2

09. SELECT grade
MAX (grade) AS max-grade,
MIN (grade) AS min-grade AND
AVG (grade) AS avg-grade
FROM enrollments-table
GROUP BY grade;
RESULTS ↓

MAX-grade	min-grade	avg-grade
90	78	84.6

10. SELECT course_id
COUNT * AS ~~total~~enrollment_count
FROM enrollments-table
GROUP BY course_id
RESULTS ↓

course_id	enrollment_count
101	1
102	2
103	3
104	4
105	5

11. SELECT department
SUM (salary) AS total_salary, AND
SUM (bonus) AS total_bonus
FROM salaries_table.
GROUP BY department;
RESULTS ↓

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70 000	6000

12. SELECT department
AVG (salary) AS avg_salary
FROM salaries_table
GROUP BY department
HAVING Avg(salary) > 55 000;
RESULTS ↓

department	avg_salary
IT	61 000

13. SELECT employees
Sum (salary + bonus) AS total_compensation
From salaries-table
Group By employees
HAVING total_compensation > 60 000
RESULTS ↓

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

14. SELECT department
Sum (budget) AS total_budget AND
AVG (budget) AS avg-budget
FROM projects-table
GROUP BY departments
HAVING avg_budget > 70 000
RESULTS ↓

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. SELECT *
FROM projects-table
WHERE budget BETWEEN 50 000 AND 120 000 AND
department != marketing;

RESULTS ↓

Project_id	project_name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000